HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Provided technical direction for a multidisciplinary Scrum team, guiding architecture, design decisions, and implementation quality for connected lighting control systems.
- Contributed hands-on to the development of **C#** software controllers and **JavaScript** configuration tools for next-generation LED drivers and lighting control systems.
- Designed and implemented connected-product capabilities using **TCP**, **UDP**, and **BLE** protocols, including device provisioning and secure communication flows.
- Mentored and coached engineers on effective AI-assisted workflows for analysis, development, testing, and documentation, ensuring independent review and validation of AI-generated outputs.
- Collaborated with Solution Architects to translate system-level vision into executable engineering plans and maintainable software designs for embedded and cloud-connected components.
- Reviewed human-written and AI-generated designs, code, and technical documentation for correctness, security, and alignment with architectural principles.
- Containerized firmware services using **Docker** and managed deployments to a cloud-based environment, ensuring efficient resource utilization and scalable operation.

SKILLS

Languages – C#, JavaScript, Embedded C, Python, Java, SQL

Frameworks & Libraries – .NET, React, Spring Boot, pandas, NumPy, scikit-learn

Protocols & Connectivity – TCP, UDP, BLE, Thread

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Kubernetes

Methodologies & Tools – Agile, Scrum, Jira, Confluence, Splunk, Dynatrace, Power BI, Tableau

PROJECTS

Embedded C Firmware for Smart Lighting Controls

- Developed embedded C firmware for next-generation lighting control networks, focusing on real-time responsiveness and low-power operation.
- Implemented secure communication protocols for device-to-cloud connectivity, ensuring data integrity and protection for connected lighting systems.
- Integrated hardware interfaces and managed firmware-update workflows, ensuring reliable and secure over-the-air updates for deployed devices.

Connected Product Configuration Tools

- Built **C#** and **JavaScript**-based configuration tools for Acuity's intelligent spaces products, enabling simplified setup and calibration for field technicians.
- Designed a user-friendly interface for managing device parameters and network settings, reducing the average configuration time by approximately **30%**.
- Developed REST APIs to integrate configuration data with backend cloud services, facilitating remote monitoring and management of deployed lighting systems.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)